

Emotion Detection & Personalized Learning Support Engine

Project Documentation

Submitted By

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Project Overview

The **Emotion Detection & Personalized Learning Support Engine** is an AI-powered educational platform designed to identify students' emotional states from free-text descriptions of their learning challenges. The system combines **BiLSTM**, **BERT Transformer**, **rule-based keyword analysis**, and **Gemini AI** to detect emotions such as **Confused**, **Curious**, **Frustrated**, **Confident**, and **Bored**. Based on the detected emotion, the platform provides personalized learning guidance, motivational support, and practical study recommendations.

The application also stores user interactions, compares the performance of multiple deep learning models, and visualizes emotion trends through an interactive Streamlit dashboard. This enables educators and learners to better understand learning difficulties, improve academic support, and enhance the overall learning experience.

Skills & Technologies Used

- Python Programming
- Responsible AI
- Generative Artificial Intelligence
- Transformer Model (BERT)
- Long Short-Term Memory (BiLSTM)
- PyTorch
- Streamlit

- Prompt Engineering
- NumPy
- Pandas
- Gemini AI
- Natural Language Processing (NLP)

Academic Year: 2026-2027

Department: computer science and engineering (Data science)

College Name: RISE KRISHNA SAI GANDHI GROUP OF INSTITUTIONS

Guide/Mentor: _____

Submitted in partial fulfillment of the requirements for the successful completion of the project on "Emotion Detection & Personalized Learning Support Engine."

Project Overview

Abstract

The **Emotion Detection & Personalized Learning Support Engine** is an AI-based educational support platform that helps identify students' emotional states from their written descriptions of learning challenges. The system leverages advanced Natural Language Processing (NLP) and Deep Learning techniques to analyze user input and classify emotions such as **Confused, Curious, Frustrated, Confident, and Bored**.

To improve prediction accuracy, the platform combines **BiLSTM (Bidirectional Long Short-Term Memory)** and **BERT (Bidirectional Encoder Representations from Transformers)** with rule-based keyword analysis. After detecting the learner's emotional state, the system uses **Gemini AI** to generate personalized learning guidance, motivational messages, and practical study recommendations.

The application also records interactions in CSV format, compares the outputs of multiple machine learning models, and presents emotion trends through an interactive Streamlit dashboard. This intelligent workflow reduces the time required to understand learner difficulties while providing consistent, supportive, and personalized academic assistance.

Problem Statement

Students often face challenges while learning complex topics but may struggle to clearly express their emotions or ask for help. Educators and teaching assistants spend valuable time interpreting student messages before providing appropriate guidance. Traditional learning systems focus only on academic performance and rarely consider the learner's emotional state.

This project addresses these challenges by automatically detecting emotions from student text and generating personalized learning support, enabling faster and more effective academic assistance.

Project Objectives

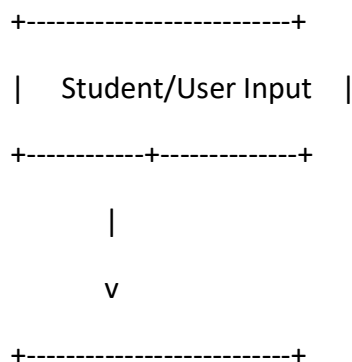
- Develop an AI-powered system to detect emotions from students' text.
- Classify emotions using both **BiLSTM** and **BERT** models.
- Improve prediction accuracy through rule-based keyword enhancement.
- Generate personalized learning guidance using **Gemini AI**.
- Compare model performance for greater transparency and reliability.
- Store interaction history for future analysis and progress tracking.
- Display emotion analytics through an interactive Streamlit dashboard.
- Support both educators and self-learning students with timely, empathetic responses.

Key Features

- AI-based emotion detection from free-text input.
- Dual-model prediction using **BiLSTM** and **BERT**.
- Mixed-emotion recognition for more accurate analysis.
- Personalized learning suggestions powered by Gemini AI.
- Rule-based keyword enhancement for improved classification.
- CSV-based interaction logging.
- Emotion trend visualization and analytics dashboard.
- User-friendly Streamlit interface for real-time interaction.

System Architecture & Technology Stack

System Architecture



| Text Preprocessing |

| (Cleaning & Tokenization) |

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| Rule-Based Keyword Enhancement Module |

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| BiLSTM Model | | BERT Model |

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| Emotion Classification |

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| Gemini AI Response |

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System Workflow

1. The learner enters a study-related problem or question.
2. The input text is cleaned and preprocessed.
3. Rule-based keyword analysis identifies important emotional cues.
4. The **BiLSTM** and **BERT** models independently predict the learner's emotion.
5. Predictions are combined to determine the final emotional state.
6. **Gemini AI** generates personalized learning guidance and motivational support.
7. The interaction is stored in a CSV file for future analysis.
8. The Streamlit dashboard displays emotion trends, confidence scores, and model comparisons.

Technology Stack

Technology	Purpose
Python	Core programming language
PyTorch	Deep learning model development
BERT	Context-aware emotion classification
BiLSTM	Sequential emotion detection
Streamlit	Interactive web application
Gemini AI	Personalized response generation
NumPy	Numerical computations
Pandas	Data preprocessing and management
Matplotlib / Seaborn	Data visualization
CSV	Interaction history storage

Skills Utilized

- Python Programming
- Responsible AI
- Generative Artificial Intelligence
- Natural Language Processing (NLP)
- Deep Learning
- Transformer Models (BERT)
- Long Short-Term Memory (BiLSTM)
- PyTorch
- Streamlit Application Development
- Prompt Engineering
- NumPy & Pandas
- Data Visualization
- Gemini AI Integration

Expected Outcome

The platform accurately identifies students' emotional states, provides personalized learning recommendations, assists educators in understanding learner needs, and improves the overall educational experience through AI-powered emotional analysis and real-time support.

Project Modules, Working Process & Use Cases

Project Modules

Module 1 – User Input Interface

- Provides a simple Streamlit-based interface.
- Allows students or educators to enter study-related text.
- Accepts free-text descriptions of learning challenges.

Module 2 – Text Preprocessing

- Cleans and normalizes the input text.
- Removes unnecessary characters and stop words.
- Tokenizes the text for machine learning models.

Module 3 – Emotion Detection

- Applies rule-based keyword analysis.
- Uses **BiLSTM** and **BERT** models to classify emotions.
- Detects emotions such as **Confused, Curious, Frustrated, Confident, and Bored**.

Module 4 – AI Response Generation

- Sends the detected emotion and user context to **Gemini AI**.
- Generates personalized learning tips, encouragement, and recommended next steps.

Module 5 – Data Storage & Analytics

- Stores user interactions in CSV format.
- Displays emotion trends and model comparison using the Streamlit dashboard.
- Supports continuous analysis and future improvements.

Working Process

Student Enters Study Problem

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Text Preprocessing

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Rule-Based Keyword Analysis

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BiLSTM & BERT Prediction

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Emotion Classification

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Gemini AI Recommendation

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Personalized Learning Support

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CSV Storage & Analytics Dashboard

Scenario 1 – Educators & Academic Support Teams

Educators and teaching assistants often spend significant time understanding students' emotions before offering academic guidance. This platform simplifies the process by automatically analyzing student messages and identifying emotions with confidence scores.

Benefits:

- Faster identification of students who need support.
- Consistent and empathetic responses.
- Reduced manual effort in sentiment interpretation.
- Comparison of BiLSTM and BERT predictions for improved reliability.
- Better academic intervention through emotion-aware insights.

Scenario 2 – Learners & Self-Study

Students learning independently can describe their academic challenges in natural language and instantly receive personalized guidance based on their emotional state.

Benefits:

- Immediate AI-powered learning support.
- Personalized study recommendations.
- Motivation and encouragement during difficult topics.
- Mixed-emotion detection for improved self-awareness.
- Progress tracking through stored interaction history.

Advantages

- Real-time emotion detection from text.
- High prediction accuracy using dual AI models.

- Personalized learning guidance with Gemini AI.
- Interactive and user-friendly Streamlit interface.
- Emotion analytics for educators and learners.
- Reduced response time and improved learning experience.
- Supports data-driven educational decision-making.

Results, Team Contributions, Future Scope & Conclusion

Results

The **Emotion Detection & Personalized Learning Support Engine** successfully demonstrates the application of Artificial Intelligence in education by identifying students' emotional states from text and providing personalized learning support. The integration of **BiLSTM**, **BERT**, **rule-based keyword analysis**, and **Gemini AI** enables accurate emotion prediction and intelligent guidance.

Key Outcomes

- Successfully detects emotions from students' textual input.
- Classifies emotions using both **BiLSTM** and **BERT** models.
- Provides personalized learning tips and motivational guidance through Gemini AI.
- Stores user interactions for future analysis and progress tracking.
- Displays emotion trends and model comparison through an interactive Streamlit dashboard.
- Reduces the time required for educators to understand and respond to students' learning difficulties.

Team Contributions

Team Member	Role	Contribution
Thallapureddy Chandra Kaladhar Reddy	Team Lead	Project planning, system architecture, model integration, and overall project coordination.
Aswardha Chandu Muttukuri	Team Member	Streamlit application development, user interface design, and dashboard implementation.
Desam Venkata Narayana	Team Member	Data preprocessing, dataset preparation, model training, and performance evaluation.

Team Member	Role	Contribution
Devireddy Ramya	Team Member	Gemini AI integration, prompt engineering, and AI response generation.
Saikasi Gosula	Team Member	Testing, documentation, analytics visualization, and deployment support.

Future Scope

The project can be enhanced with additional features to improve learning support and user experience.

- Voice-based emotion detection using speech analysis.
- Facial expression recognition for multimodal emotion detection.
- Mobile application for Android and iOS platforms.
- Support for multiple regional and international languages.
- Cloud deployment for improved scalability and accessibility.
- Integration with Learning Management Systems (LMS).
- Personalized study plan recommendations based on learning history.
- Advanced analytics dashboard for educators and institutions.

Conclusion

The **Emotion Detection & Personalized Learning Support Engine** is an innovative AI-powered educational platform that bridges the gap between emotional understanding and personalized learning. By combining advanced Natural Language Processing, Deep Learning, Transformer models, and Generative AI, the system delivers accurate emotion detection and tailored academic guidance in real time.

This platform benefits both educators and learners by reducing manual effort, improving communication, and providing timely, empathetic support. Its scalable architecture, intelligent analytics, and personalized recommendations make it a valuable solution for modern digital education environments.

References

1. BERT: Bidirectional Encoder Representations from Transformers (Devlin et al.)
2. PyTorch Official Documentation

3. Streamlit Documentation
4. Google Gemini AI Documentation
5. Python Official Documentation
6. NumPy & Pandas Documentation
7. Research papers on Emotion Detection using Natural Language Processing (NLP).